HOSPITAL CARDIOVASCULAR AI IMAGING CENTER

Epicardial & Pericoronary Fat AI Quantification Report

Automated 3D WholeBody SegResNet & CRISP-CT Coronary Inflammation Analysis

Patient ID: 0987
Exam Date: 2026-10-29
Modality: Cardiac CT (CCTA)

LOW RISK

CRISP-CT Vascular Inflammation & Fat Burden: Normal (~25th %ile)

Threshold: PVAT > -70.1 HU indicates active inflammation.

EPICARDIAL ADIPOSE TISSUE (EAT)

Total EAT Volume: 50.3 cm³

Mean Attenuation: -73.4 HU

Chamber Sub-Compartments:

- Left Ventricle (LV): 18.6 cm³
- Right Ventricle (RV): 30.0 cm³
- Left Atrium (LA): 11.4 cm³
- Right Atrium (RA): 13.6 cm³

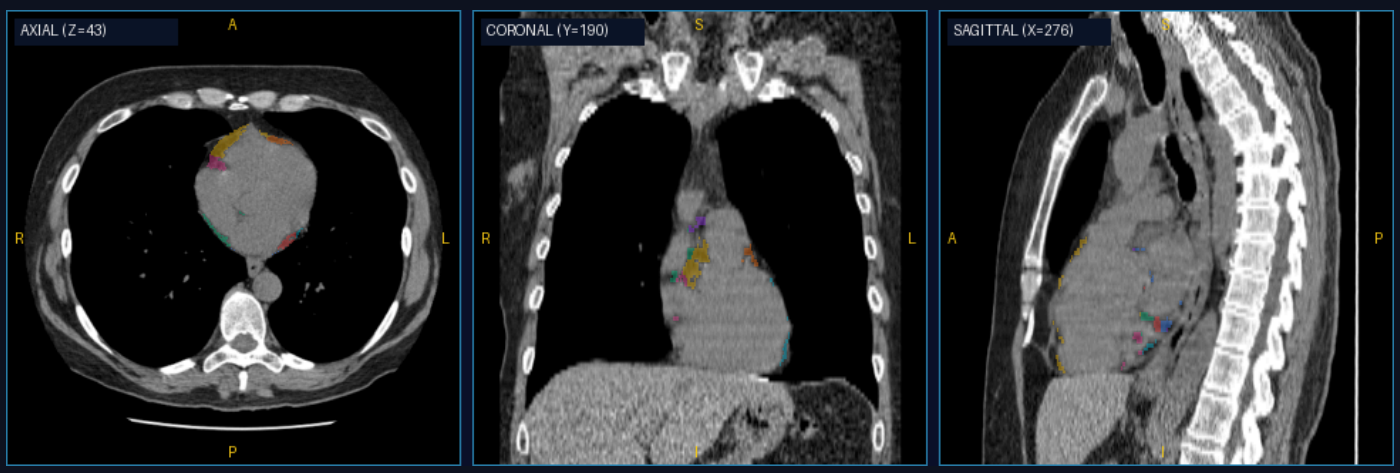
PERICORONARY ADIPOSE TISSUE (PVAT)

LAD PVAT:	6.0 cm³	-72.8 HU	Normal
LCX PVAT:	5.4 cm³	-72.1 HU	Normal
RCA PVAT:	5.3 cm³	-72.4 HU	Normal

Total Pericoronary Fat Volume: 16.6 cm³

3-VIEW ORTHOGONAL MULTI-PLANAR RECONSTRUCTION (MPR) WITH AI OVERLAYS

Axial (Z), Coronal (Y), and Sagittal (X) cardiac planes with 8-compartment adipose masks



STRUCTURED CLINICAL IMPRESSION (READY FOR EHR / POWERSCRIBE):

```
QUANTITATIVE CARDIAC ADIPOSE & INFLAMMATION AI ANALYSIS:
- Overall Biomarker Triage: LOW RISK
- Total Epicardial Fat (EAT) Volume: 50.3 cm3 (Normal burden, ~25th %ile)
- Total EAT Mean Attenuation: -73.4 HU
- Pericoronary Fat Attenuation (CRISP-CT active inflammation threshold: > -70.1 HU):
    * LAD PVAT: -72.8 HU - Normal Attenuation (· -70.1 HU)
    * LCX PVAT: -72.1 HU - Normal Attenuation (· -70.1 HU)
    * RCA PVAT: -72.4 HU - Normal Attenuation (· -70.1 HU)
IMPRESSION: Total Epicardial Adipose Tissue (EAT) volume is 50.3 cm³ (Normal burden, ~25th percentile), with mean attenuation
```


For Investigational / Research Use Only (RUO). Clinical verification required by board-certified physician.

Patient ID: 0987
Exam Date: 2026-10-29

AI GENERATIVE NARRATIVE FINDINGS & CLINICAL IMPRESSION:

TECHNIQUE: Non-contrast CT.

FINDINGS:
Medical devices: None present.

CHEST:
Lungs and airways: The lungs are clear with no infiltrate, nodule, or interstitial disease. Central airways are patent.
Pleura: No pleural effusion or pneumothorax.
Mediastinum and hila: No bulky mediastinal adenopathy is seen.
Heart and pericardium: Normal heart size. Minimal pericardial fluid.
Great vessels: The great vessels are unremarkable.
Esophagus: No esophageal abnormality.
Chest wall and breast: No chest wall or breast mass.
Bones: Visualized osseous structures are normal.

IMPRESSION:

1. Minimal pericardial fluid.
<answer> Pericardial effusion </answer>

For Investigational / Research Use Only (RUO). Clinical verification required by board-certified physician.